

8.1 Identifiability

Section 6.4 ended with a table of four coefficient vectors that disagreed in every entry, some even in sign, and still produced the same fitted values. Nothing had gone wrong with the computation. The model matrix had linearly dependent columns, and so the least squares problem had a whole affine subspace of solutions. This chapter asks what can be learned about β in that situation. We begin with a question that comes before estimation and does not involve data at all. Suppose we knew the distribution of $\mathbf{Y}$ exactly, as if we had infinitely many replications. Would we then know β ? If not, which features of β would we know?

8.1.1 What the distribution of the data can reveal

Throughout, $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$ is the linear model of Definition 5.3.1: $\mathbf{X}$ is a fixed $n \times p$ matrix of rank r , and $E(\varepsilon) = \mathbf{0}$. We do not assume $r = p$. The coefficient vector enters the model only through the product $\mathbf{X}\beta$. Suppose the distribution of the error vector ε is allowed to depend on other parameters, such as σ^2 , but not on β . Then the distribution of $\mathbf{Y}$ is the distribution of ε shifted by the vector $\mathbf{X}\beta$. Two coefficient vectors with the same product $\mathbf{X}\beta$ therefore give the same distribution of $\mathbf{Y}$, with the other parameters held fixed. Conversely, the distribution of $\mathbf{Y}$ determines its mean $E(\mathbf{Y}) = \mathbf{X}\beta$. So the most we can learn about β from the distribution is the value of $\mathbf{X}\beta$. The mean vector

$$\mu = \mathbf{X}\beta \in \mathcal{C}(\mathbf{X})$$

is the object the data speak about. A value of β is one way of writing μ in coordinates.

This observation reduces the question to linear algebra. The map $\beta \mapsto \mathbf{X}\beta$ from $\mathbb{R}^p$ to $\mathbb{R}^n$ is linear. Two coefficient vectors have the same image iff their difference lies in the null space $\mathcal{N}(\mathbf{X})$. So

$$\{\mathbf{b} \in \mathbb{R}^p : \mathbf{X}\mathbf{b} = \mathbf{X}\beta\} = \beta + \mathcal{N}(\mathbf{X}), \quad (8.1.1)$$

an affine subspace of dimension $p - r$ by rank-nullity (Eq. 6.2.1). All vectors in Eq. 8.1.1 are observationally equivalent: no experiment with this model matrix can distinguish between them.

Definition 8.1.1 — Identifiability

In the linear model $E(\mathbf{Y}) = \mathbf{X}\beta$, $\beta \in \mathbb{R}^p$, the parameter β is **identifiable** if $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$ implies $\beta_1 = \beta_2$. A function g defined on $\mathbb{R}^p$, with values in any set, is **identifiable** if $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$ implies $g(\beta_1) = g(\beta_2)$.

The definition is phrased through the mean, not the whole distribution. By the argument above the two versions agree whenever the error distribution does not involve β , which is the situation throughout this book until the generalized linear models of Part VIII. There the mean is $h(\mathbf{X}\beta)$ for a known one-to-one function h , and the same definition applies unchanged. It says that $g(\beta)$ is a well-defined function of the point μ , whichever coordinates were used to describe that point.

Theorem 8.1.1 — Identifiable means “a function of the mean”

A function g on $\mathbb{R}^p$ is identifiable iff there is a function h on $\mathcal{C}(\mathbf{X})$ with

$$g(\beta) = h(\mathbf{X}\beta) \quad \text{for every } \beta \in \mathbb{R}^p.$$

Equivalently, g is constant on each affine subspace $\beta + \mathcal{N}(\mathbf{X})$.

Proof. If $g = h \circ \mathbf{X}$ and $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$, then $g(\beta_1) = h(\mathbf{X}\beta_1) = h(\mathbf{X}\beta_2) = g(\beta_2)$. Conversely, let g be identifiable. For $\mu \in \mathcal{C}(\mathbf{X})$ choose any β with $\mathbf{X}\beta = \mu$ and put $h(\mu) = g(\beta)$. The value does not depend on the choice, because any two choices have the same image under $\mathbf{X}$. By construction $g(\beta) = h(\mathbf{X}\beta)$. The last sentence restates the definition using Eq. 8.1.1. ■

The theorem makes identifiability a statement about information. A function that can be computed from μ is one that could, in principle, be estimated consistently, provided that μ itself can be estimated. A function that cannot be computed from μ is one about which the model is silent. Its value can be anything along the affine subspace, and more data from the same design cannot help.

8.1.2 When the parameter is identifiable**Proposition 8.1.2**

The following are equivalent: (a) β is identifiable; (b) $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$; (c) $\text{rank}(\mathbf{X}) = p$; (d) $\mathbf{X}^\top \mathbf{X}$ is nonsingular. In that case the function h of Theorem 8.1.1 for $g(\beta) = \beta$ is $h(\mu) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mu$.

Proof. By Eq. 8.1.1, β is identifiable iff every affine subspace $\beta + \mathcal{N}(\mathbf{X})$ is a single point, which is (b). (b) $\Leftrightarrow$ (c) is rank-nullity, and (c) $\Leftrightarrow$ (d) is Corollary 6.2.4(c). If $\mathbf{X}^\top \mathbf{X}$ is nonsingular and $\mu = \mathbf{X}\beta$, then $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mu = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\beta = \beta$. ■

When $\text{rank}(\mathbf{X}) = r < p$ the parameter is not identifiable, but parts of it may be. The simplest parts are the individual coordinates.

Proposition 8.1.3 — Identifiability of a single coefficient

Write $\mathbf{x}_1, \dots, \mathbf{x}_p$ for the columns of $\mathbf{X}$. For each j the following are equivalent:

- (a) β_j is identifiable;
- (b) every $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ has $v_j = 0$;

- (c) $\mathbf{x}_j$ is not a linear combination of the other columns of $\mathbf{X}$;
- (d) deleting column j from $\mathbf{X}$ lowers the rank.

Proof. (a) $\Leftrightarrow$ (b): by Eq. 8.1.1, the j th coordinates of the vectors observationally equivalent to $\boldsymbol{\beta}$ are $\beta_j + v_j$, $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. They all equal β_j iff $v_j = 0$ for every null vector. (b) $\Leftrightarrow$ (c): a relation $\mathbf{x}_j = \sum_{k \neq j} a_k \mathbf{x}_k$ is the same as a null vector with j th entry -1 , and any null vector with $v_j \neq 0$ can be rescaled to have $v_j = -1$. (c) $\Leftrightarrow$ (d): the column space loses a dimension when $\mathbf{x}_j$ is removed iff $\mathbf{x}_j$ is not in the span of the remaining columns. ■

Part (c) gives a practical test. A coefficient is lost exactly when its regressor is *aliased*, that is, when it can be rebuilt from the others. In a model that is only partly identifiable, some coefficients can still be perfectly well identified. If $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}_j]$ and $\mathbf{x}_j \notin \mathcal{C}(\mathbf{X}_1)$, then β_j is identifiable however badly $\mathbf{X}_1$ is parameterized. A typical case is a numeric covariate added to an overparameterized factor model: the covariate's coefficient is identifiable as long as the covariate is not constant within every level of the factor (Section 8.6).

8.1.3 Where rank deficiency comes from

Linear dependence among the columns of $\mathbf{X}$ arises in three quite different ways, and it helps to distinguish them.

By construction. A factor with g levels entered as an intercept together with all g indicator columns gives $g+1$ columns spanning a g -dimensional space (Example (A rank-deficient layout)). Such *overparameterized* models are written down deliberately, because they treat the levels symmetrically. The dependence is harmless once it is understood, and most of this chapter is about understanding it.

By the logic of the variables. Some sets of variables satisfy an identity that holds for every observation. Examples are shares of a whole that add to one, a total entered together with all its parts, and, the best-known case, age, period and birth cohort.

By the design. Two regressors that are logically unrelated may be exactly collinear in the sample at hand. A factor level may never be observed. The design may have fewer distinct rows than parameters. In each case another data set with the same model could identify the parameter, but this one cannot.

Identifiability is therefore a property of the model matrix, and through it of the design, not of the population being studied. The next example shows the second kind of dependence.

Example. (Age, period and cohort)

A survey is fielded every five years from 2000 to 2020, and its respondents are grouped in eleven five-year age bands from 20 to 70. For each of the 55 combinations we have one average response. A demographer wants to separate three linear trends: in age (people change as they grow older), in period (everyone changes with the times), and in birth cohort (generations differ). With age, period and cohort centred at 45, 2010 and 1965, the model is

$$E(y) = \beta_0 + \beta_A \text{age} + \beta_P \text{period} + \beta_C \text{cohort}.$$

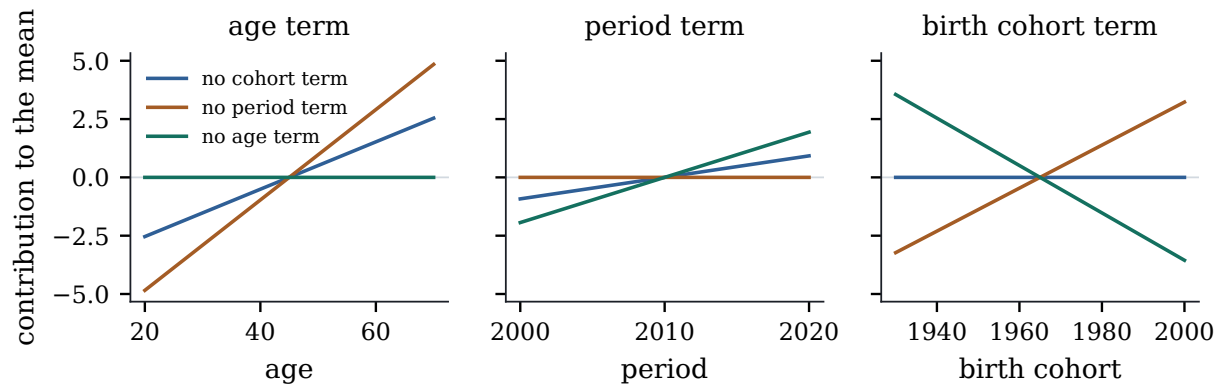


Figure 8.1.1: Three equally good fits of the age–period–cohort model. Each panel shows the contribution of one term to the fitted mean under the three fits of [Example \(Age, period and cohort\)](#). The sum of the three contributions is the same for every respondent, so the data cannot choose between the fits.

But cohort is period minus age, for every respondent. The four columns satisfy age – period + cohort = $\mathbf{0}$, so $\mathbf{v} = (0, 1, -1, 1)^\top$ spans $\mathcal{N}(\mathbf{X})$ and $\text{rank}(\mathbf{X}) = 3$. By [Proposition 8.1.3](#), none of the three slopes is identifiable.

The listing fits the model three times, each time dropping one of the three columns. All three fits have residual sum of squares 157.00, because they have the same column space. They tell three different stories:

Fit	age slope	period slope	cohort slope
no cohort term	0.102	0.092	0.000
no period term	0.194	0.000	0.092
no age term	0.000	0.194	-0.102

One analyst reports a steady ageing effect and no generational change. Another reports a strong ageing effect and a positive cohort trend. A third reports no ageing at all, a strong secular trend, and younger generations scoring lower. The data were in fact simulated with slopes 0.30, -0.10 and 0.20. That vector lies on the parallel line $\boldsymbol{\beta} + t\mathbf{v}$ of equally valid truths, and the data cannot prefer it. Only the identifiable combinations are recovered: the fitted 0.194 and 0.092 estimate the true 0.20 and 0.10.

What the three fits share are the linear functions orthogonal to $\mathbf{v} = (0, 1, -1, 1)^\top$:

$$\beta_A + \beta_P = 0.194, \quad \beta_P + \beta_C = 0.092, \quad \beta_A - \beta_C = 0.102,$$

in every row of the table, and β_0 . Any statement about the separate trends requires information from outside these data, for instance a constraint justified by subject knowledge, and the conclusions then rest on that constraint ([Exercise \(C1\)](#)). [Figure 8.1.1](#) shows how each fit divides the same fitted surface among the three terms.

```
import numpy as np

ages = np.arange(20, 75, 5)           # 11 age bands: 20, 25, ..., 70
periods = np.arange(2000, 2025, 5)   # 5 survey waves: 2000, ..., 2020
A, P = np.meshgrid(ages, periods, indexing="ij")
age, period = A.ravel() - 45.0, P.ravel() - 2010.0 # centred at 45 and 2010
```

```

cohort = period - age                                # birth year minus 1965, exactly
X = np.column_stack([np.ones(age.size), age, period, cohort])
print("n =", X.shape[0], " p =", X.shape[1], " rank =", np.linalg.matrix_rank(X))
print("X @ (0, 1, -1, 1) =", np.abs(X @ [0, 1, -1, 1]).max())

```

[Run this code online](#)

```

rng = np.random.default_rng(8)
truth = np.array([50.0, 0.30, -0.10, 0.20])
y = X @ truth + rng.normal(scale=1.5, size=age.size)

def fit_without(j):
    """Least squares after deleting column j (one of age, period, cohort)."""
    keep = [k for k in range(4) if k != j]
    b = np.zeros(4)
    b[keep] = np.linalg.lstsq(X[:, keep], y, rcond=None)[0]
    return b

stories = {"no cohort term": fit_without(3),
           "no period term": fit_without(2),
           "no age term": fit_without(1)}
for name, b in stories.items():
    sse = np.sum((y - X @ b) ** 2)
    print(f"{name:15s} b = {np.round(b, 3)} SSE = {sse:.3f}",
          f" age+period = {b[1] + b[2]:.3f} period+cohort = {b[2] + b[3]:.3f}")

```

[Run this code online](#)

8.1.4 Linear and nonlinear identifiable functions

In the example the identifiable linear functions were exactly those whose coefficient vectors are orthogonal to the null space. That is true in general.

Proposition 8.1.4

A linear function $\lambda^\top \beta$ is identifiable iff $\lambda \perp \mathcal{N}(\mathbf{X})$, that is, iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. The identifiable linear functions form a subspace of dimension $r = \text{rank}(\mathbf{X})$.

Proof. By [Theorem 8.1.1](#), $\lambda^\top \beta$ is identifiable iff $\lambda^\top (\beta + \mathbf{v}) = \lambda^\top \beta$ for all β and all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, that is, iff $\lambda^\top \mathbf{v} = 0$ for all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. By [Corollary 6.2.4\(d\)](#), $\mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$, which has dimension $\text{rank}(\mathbf{X}^\top) = r$. ■

Identifiable functions need not be linear. If $\lambda_1^\top \beta$ and $\lambda_2^\top \beta$ are identifiable, so are their ratio (where defined), their product, and $\max\{\lambda_1^\top \beta, 0\}$. Every function of μ is identifiable, for example $\|\mu\|^2$ or the largest mean. For instance, in a quadratic regression $E(y) = \beta_0 + \beta_1 x + \beta_2 x^2$ with a full-rank design, the location $-\beta_1/(2\beta_2)$ of the turning point is identifiable. It is not linear, and Chapter 12 has to estimate it with care. The next section shows that among the *linear* functions, identifiability coincides with a much more concrete property, the existence of a linear unbiased estimator.

Idea. The data determine the mean vector $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ and nothing else about $\boldsymbol{\beta}$. A function of $\boldsymbol{\beta}$ is identifiable exactly when it is a function of $\boldsymbol{\mu}$. A linear function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is identifiable exactly when $\boldsymbol{\lambda}$ lies in the row space $\mathcal{C}(\mathbf{X}^\top)$, the orthogonal complement of the invisible directions $\mathcal{N}(\mathbf{X})$.

8.1.5 Exercises

8.1.5.1 A. Check your understanding

Exercise A1.

In [Example \(Age, period and cohort\)](#), show directly that β_0 , $\beta_A + \beta_P$ and $\beta_P + \beta_C$ are identifiable, that every identifiable linear function is a combination of these three, and that $\beta_A - \beta_C$ is one of them. Is $\beta_A + \beta_C$ identifiable?

Exercise A2.

A household's spending is split into food, housing and other, with shares $s_1 + s_2 + s_3 = 1$. A regression of savings on an intercept and all three shares is proposed. Find $\mathcal{N}(\mathbf{X})$ (assuming the shares vary enough that there is no other relation among the columns), decide which coefficients are identifiable, and describe two standard ways to make the model identifiable.

8.1.5.2 B. Practice

Exercise B1.

Let $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_g]$ be the indicator matrix of a factor and $\mathbf{x}$ a numeric covariate, and consider $\mathbf{X} = [\mathbf{1}, \mathbf{Z}, \mathbf{x}]$. Show that the slope of $\mathbf{x}$ is identifiable iff $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$, that is, iff $\mathbf{x}$ is not constant within every level. Show that no intercept-type parameter β_0 or α_k is identifiable.

Solution. The columns $\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g$ satisfy $\mathbf{1} - \sum_k \mathbf{z}_k = \mathbf{0}$, so they are dependent, and $\mathcal{C}([\mathbf{1}, \mathbf{Z}]) = \mathcal{C}(\mathbf{Z})$. By [Proposition 8.1.3\(c\)](#), the slope is identifiable iff $\mathbf{x}$ is not a combination of $\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g$, that is, iff $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$. A vector lies in $\mathcal{C}(\mathbf{Z})$ iff it is constant within each level. For the other coefficients, $\mathbf{v} = (1, -1, \dots, -1, 0)^\top$ is a null vector with nonzero entries in positions $0, 1, \dots, g$, so by [Proposition 8.1.3\(b\)](#) none of $\beta_0, \alpha_1, \dots, \alpha_g$ is identifiable.

Exercise B2.

Let $\mathbf{P}$ be the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$. Show that g is identifiable iff $g(\boldsymbol{\beta}) = g(\mathbf{P}\boldsymbol{\beta})$ for every $\boldsymbol{\beta}$. Deduce that the identifiable functions are exactly the functions of $\mathbf{P}\boldsymbol{\beta}$, and that $\mathbf{P}\boldsymbol{\beta}$ is the unique coefficient vector in $\mathcal{C}(\mathbf{X}^\top)$ representing the mean $\mathbf{X}\boldsymbol{\beta}$.

Exercise B3.

A design with model matrix $\mathbf{X}$ of rank $r < p$ is augmented by one more observation at a point $\mathbf{x}_0$. Show that the rank increases iff $\mathbf{x}_0 \notin \mathcal{C}(\mathbf{X}^\top)$. Show that, in the age–period–cohort design, no additional observation can make the slopes identifiable, and explain why in words.

8.1.5.3 C. Going deeper**Exercise C1.**

Replace the three linear terms of [Example \(Age, period and cohort\)](#) by three factors: an indicator for each age band, each period and each cohort (with an intercept). Show that the null space of the resulting model matrix has dimension 3 plus exactly one extra dimension. Identify the extra null vector: it assigns the linear sequences $a - \bar{a}$, $-(t - \bar{t})$ and $c - \bar{c}$ to the age, period and cohort effects. Deduce that every second difference of age effects, such as $\alpha_{a+1} - 2\alpha_a + \alpha_{a-1}$, is identifiable, while no linear trend in any single one of the three factors is.

Solution. Each factor's indicators sum to $\mathbf{1}$, which gives the usual three null vectors (intercept minus all indicators of one factor). The identity age – period + cohort = const holds for every observation. Writing the age of an observation as $\sum_a a z_a^A$, and similarly for period and cohort, it becomes a relation $\sum_a a z_a^A - \sum_t t z_t^P + \sum_c c z_c^C - \kappa \mathbf{1} = \mathbf{0}$ among the columns. Subtracting suitable multiples of the three standard relations centres the three sequences, and this gives the stated vector. It is not a combination of the other three, because those are constant on each factor's effects and this one is linear and nonconstant. To see that there are no more, check that the matrix has rank $1 + (A - 1) + (P - 1) + (C - 1) - 1$. This rank check can be done directly or numerically for the design of [Example \(Age, period and cohort\)](#). A function $\sum_a d_a \alpha_a$ is identifiable iff $\mathbf{d}$ is orthogonal to all four null vectors, that is, iff $\sum_a d_a = 0$ and $\sum_a d_a a = 0$. A second difference satisfies both. A linear trend $\sum_a (a - \bar{a}) \alpha_a$ fails the second condition.